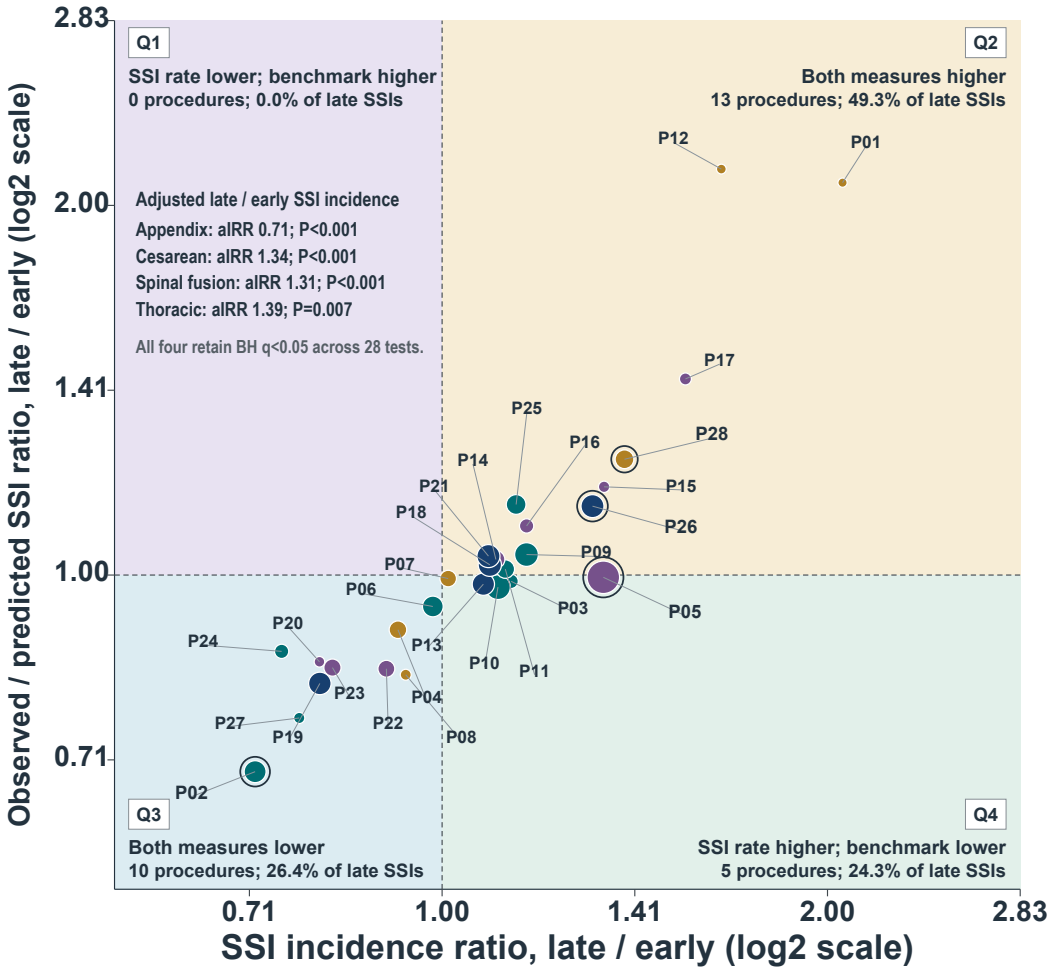
a. Profile similarity

b. Procedure evidence and calendar-year observed/predicted profile

Procedure	Operations	SSIs / 1,000	Observed / predicted	Signal reports, %	Next signal aRR (95% CI), BH q	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Coronary bypass, chest only	16,877	4.9	0.70	6.7	1.08 (0.53-2.20); =0.931										
Aortic aneurysm repair	3,487	4.6	0.67	1.8	NE										
Rectal	64,016	6.1	0.39	6.1	1.55 (0.96-2.49); =0.089										
Liver transplant	9,878	26.6	0.55	17.3	NE	*		*							*
Kidney transplant	25,247	6.1	0.96	19.2	1.81 (1.09-3.01); =0.030									*	*
Exploratory laparotomy	452,467	6.0	1.02	20.7	1.69 (1.48-1.92); <0.001									*	*
Cesarean	1,200,547	1.7	0.92	18.7	1.96 (1.66-2.30); <0.001									*	*
Knee prosthesis	496,006	3.5	0.92	25.0	1.87 (1.63-2.16); <0.001		*	*	*			*	*		
Hip prosthesis	415,988	5.8	0.91	27.0	1.47 (1.28-1.68); <0.001	*	*	*	*		*	*	*	*	*
Spinal fusion	410,369	7.4	0.91	25.1	2.10 (1.78-2.48); <0.001				*			*		*	*
Fracture reduction	400,655	6.7	0.94	19.1	1.51 (1.32-1.73); <0.001										*
Gallbladder	507,649	3.4	0.88	17.5	1.57 (1.35-1.82); <0.001										
Laminectomy	342,895	2.7	0.79	17.7	1.83 (1.50-2.22); <0.001										*
Abdominal hysterectomy	204,922	6.1	0.83	19.9	1.58 (1.34-1.86); <0.001										
Coronary bypass, chest + donor	131,623	5.9	0.80	27.3	1.61 (1.27-2.03); <0.001	*		*	*		*	*		*	*
Gastric	202,535	5.1	0.69	16.5	1.55 (1.26-1.91); <0.001										
Appendix	378,956	2.6	0.68	11.7	1.54 (1.29-1.84); <0.001										
Heart transplant	4,671	6.9	0.56	15.0	NE								*		*
Thoracic	206,270	2.9	0.79	9.2	2.09 (1.63-2.69); <0.001										
Cardiac	156,011	3.5	0.76	12.4	1.72 (1.33-2.23); <0.001										
Kidney	73,759	3.4	0.99	9.2	1.31 (0.85-2.02); =0.274										
Vaginal hysterectomy	42,259	4.6	0.88	7.1	1.50 (1.02-2.23); =0.055										
Spleen	18,043	6.7	1.09	5.5	2.11 (1.24-3.58); =0.008										
Ovarian	178,558	0.7	0.95	3.9	1.11 (0.49-2.49); =0.931										
Pacemaker	172,140	1.8	1.06	9.5	2.83 (2.07-3.87); <0.001										
Small bowel	245,423	16.7	0.84	24.8	1.69 (1.48-1.93); <0.001										
Colon	295,932	22.7	0.89	27.5	1.76 (1.53-2.02); <0.001	*	*	*				*	*	*	*
Bile, liver, pancreas	80,789	28.8	1.01	16.7	2.57 (2.06-3.21); <0.001										

Observed / predicted SSI ratio: <=0.75 >0.75 to 1.00 >1.00 to 1.25 >1.25

c. Direction, burden and statistical evidence



Bone and spine
Digestive procedures
Heart, chest and aorta
Other procedures

Ring: adjusted incidence change with BH q<0.05

Bubble size: late operations; codes identify the procedures in panel d.

d. Adjusted late / early incidence comparisons

Procedure	aIRR (95% CI)	P	BH q
P01 Aortic aneurysm repair	2.05 (0.77-5.51)	0.153	0.326
P02 Appendix	0.71 (0.60-0.84)	<0.001	0.001
P03 Bile, liver, pancreas	1.12 (0.96-1.32)	0.159	0.326
P04 Cardiac	0.95 (0.75-1.19)	0.645	0.785
P05 Cesarean	1.34 (1.14-1.58)	<0.001	0.003
P06 Colon	0.98 (0.90-1.08)	0.699	0.815
P07 Coronary bypass, chest + donor	1.01 (0.83-1.24)	0.915	0.985
P08 Coronary bypass, chest only	0.93 (0.56-1.54)	0.786	0.881
P09 Exploratory laparotomy	1.16 (1.00-1.35)	0.047	0.148
P10 Gallbladder	1.10 (0.97-1.25)	0.141	0.326
P11 Gastric	1.11 (0.92-1.35)	0.283	0.431
P12 Heart transplant	NE	NE	NE
P13 Hip prosthesis	1.08 (0.93-1.25)	0.323	0.431
P14 Abdominal hysterectomy	1.10 (0.95-1.28)	0.194	0.340
P15 Vaginal hysterectomy	1.33 (0.93-1.90)	0.120	0.326
P16 Kidney	1.16 (0.87-1.56)	0.309	0.431
P17 Kidney transplant	1.56 (1.03-2.34)	0.035	0.122
P18 Knee prosthesis	1.09 (0.92-1.29)	0.304	0.431
P19 Laminectomy	0.81 (0.68-0.97)	0.024	0.095
P20 Liver transplant	NE	NE	NE
P21 Fracture reduction	1.08 (0.96-1.22)	0.184	0.340
P22 Ovarian	0.90 (0.61-1.33)	0.598	0.761
P23 Pacemaker	0.81 (0.61-1.09)	0.163	0.326
P24 Rectal	0.74 (0.58-0.95)	0.020	0.092
P25 Small bowel	1.14 (1.02-1.27)	0.019	0.092
P26 Spinal fusion	1.31 (1.15-1.49)	<0.001	0.001
P27 Spleen	0.77 (0.51-1.18)	0.228	0.375
P28 Thoracic	1.39 (1.10-1.77)	0.007	0.048